

Theme Park · Amenities — Functions Calling Functions — English Worksheet (Beginner)

Topic: Amenities — Functions Calling Functions · **Course:** Theme Park Creator · **Level:** Beginner ·

Time: about 30 minutes

Your park has stalls. Now it needs **amenities** — a **bathroom**, a **janitor room**.

Both are rooms. So you write **one** room function, and your amenity functions **call** it.

These are the blocks you use this lesson:

```
blocks.fill(QUARTZ_BLOCK, pos(0, 0, 0), pos(4, 3, 4), FillOperation.HOLLOW)
blocks.place(CAULDRON, pos(1, 1, 1))
```

`blocks.fill(...)` fills a box between two corners. `FillOperation.HOLLOW` builds only the **walls** and leaves the middle empty. `blocks.place(block, pos(x, y, z))` puts **one** block in **one** spot. `CAULDRON` = the toilet. `BARREL` = the janitor's supplies.

1 · Meet Helper Functions

A **helper function** does one job. Other functions **call** it instead of writing that job again.

```
build_bathroom(5)
├──► build_room(QUARTZ_BLOCK, 5)  ← the helper builds the walls
└──► place CAULDRON              ← the bathroom adds its own thing
```

```
def build_room(wall, size):
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)

def build_bathroom(size):
    build_room(QUARTZ_BLOCK, size)
    blocks.place(CAULDRON, pos(1, 1, 1))

build_bathroom(5)
```

Which function builds the walls? Circle one: `build_room` · `build_bathroom` · `blocks.place`

Which function places the cauldron? Circle one: `build_room` · `build_bathroom` · `blocks.fill`

2 · Predict 🧙‍♂️

Read the code. Circle your answer.

```
def build_room(wall, size):  
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)  
  
def build_bathroom(size):  
    build_room(QUARTZ_BLOCK, size)  
    blocks.place(CAULDRON, pos(1, 1, 1))  
  
build_bathroom(5)
```

`build_bathroom(5)` sends `5` to the helper. How wide is the room? Circle one: `5` · `3` · `1`

What are the walls made of? Circle one: `QUARTZ_BLOCK` · `CAULDRON` · `BARREL`

```
def build_janitor_room(size):  
    build_room(OAK_PLANKS, size)  
    blocks.place(BARREL, pos(1, 1, 1))  
  
build_janitor_room(4)
```

The same helper builds this room too. What are these walls made of? Circle one: `OAK_PLANKS` · `QUARTZ_BLOCK` · `BARREL`

3 · Find the Difference 🐉

Clean code first, then a broken version. Circle the answer.

Pair A — The bathroom should have walls **and** a cauldron.

```
# clean
def build_bathroom(size):
    build_room(QUARTZ_BLOCK, size)
    blocks.place(CAULDRON, pos(1, 1, 1))
```

```
# buggy
def build_bathroom(size):
    blocks.place(CAULDRON, pos(1, 1, 1))
```

What is wrong? Circle one: it never calls `build_room` · it never places the cauldron · nothing

Pair B — The helper wants the material first, then the size.

```
# clean
def build_bathroom(size):
    build_room(QUARTZ_BLOCK, size)
```

```
# buggy
def build_bathroom(size):
    build_room(size, QUARTZ_BLOCK)
```

What is wrong? Circle one: the two arguments are swapped · the size is missing · the material is missing

4 · Fill the Gap

The janitor room should have wooden walls, then a barrel. One line is missing.

```
def build_room(wall, size):  
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)  
  
def build_janitor_room(size):  
    _____  
    blocks.place(BARREL, pos(1, 1, 1))  
  
build_janitor_room(4)
```

Which line is missing? Circle one: `build_room(OAK_PLANKS, size)` · `build_janitor_room(4)` · `blocks.place(BARREL, pos(1, 1, 1))`

5 · Show Your Work 📷 🎥

Switch to your homework world. Write a `build_room` helper, then a `build_bathroom` function that **calls** it and places a `CAULDRON`. Call `build_bathroom` once.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.